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# Sensitivity Characteristics of Horizontal and Vertical Hall Sensors in the Voltage- and Current-Mode

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**Abstract**—This paper analyses sensitivity of Hall devices in the horizontal and vertical forms operating in the current and voltage-mode. The 2D finite element model (FEM) implemented by COMSOL Multiphysics evaluates sensitivity efficiencies. Different geometries of horizontal and vertical Hall sensor plates have been studied and simulated. Among them, a twin cross-shaped coupled horizontal Hall device and a four-folded three contacts vertical Hall device model displayed the best sensitivity. Simulations results of sensitivity performance, obtained in two different horizontal and vertical shapes, are compared and discussed with state-of-the-art. Simulations show that the current-mode operation shows sensitivity better than  $11\%T^{-1}$  and  $16\%T^{-1}$  for cross-shaped horizontal Hall device and four-folded 3-contact vertical Hall device, respectively.

## I. INTRODUCTION

Hall effect sensors have been the workhorse magnetic sensor for decades. Their performance, size and low cost have made them the most popular type of magnetic devices. This is even more true since Hall sensors are easily integrated in modern commercial CMOS technologies.

The integrated Hall sensors devices can be build up with different geometry and sense contacts, and basing on that they can be classified as horizontal or vertical. The former, also known as Hall plate, has an horizontal plate parallel to the semiconductor substrate and the current lines are parallel to the chip surface, so that it is sensitive to the magnetic field,  $B_z$ , perpendicular to the sensor plate surface, as shown in Fig. 1. The active region of the device is compatible with the layers available in CMOS technologies. The area of integrated horizontal Hall devices is on the order of tens of  $\mu m^2$ . Their current-biased voltage-mode sensitivities ( $S_{V_I}$ ) range from  $150 \sim 400 VA^{-1}T^{-1}$ , while typical voltage-biased voltage-mode sensitivities ( $S_{V_V}$ ) are  $2.5 \sim 5\%T^{-1}$ .

On the other side, vertical Hall sensors (VHS), of whose Fig. 6 shows a possible cross section, are capable of measuring surface-parallel components of the magnetic field,  $B_y$ . They enable simple single-chip multi-axial magnetic sensors. The vertical Hall sensors in the current-biased voltage-mode has sensitivities ( $S_{V_I}$ ) in the range from  $100 \sim 400 VA^{-1}T^{-1}$ , while typical voltage-biased voltage-mode sensitivities ( $S_{V_V}$ ) are  $1 \sim 4\%T^{-1}$ .

The result obtained by this paper show that sensitivities in the current-mode condition are better than the voltage-mode. Simulation show a sensitivity of  $11\%T^{-1}$  for the horizontal device and as high as  $16\%T^{-1}$  for the vertical device.

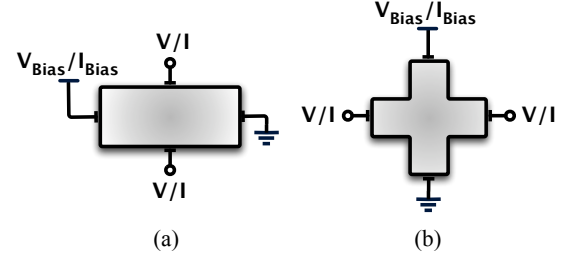


Fig. 1. Hall sensor (a) conventional rectangular plate and (b) symmetric cross-shaped plate suitable for current spinning.

The paper is organized as follows: in the next Section, the definition of sensitivity of current-mode and voltage-mode magnetic Hall sensors are recalled. Then, a performance comparison is discussed. Section IV shows the simulations in 2D COMSOL of a twin cross-shaped coupled horizontal Hall device. Simulations of the four-folded 3-contact vertical Hall device are also presented in Section V. Section VI draws the conclusions.

## II. SENSITIVITY PERFORMANCE IN MAGNETIC HALL SENSORS

Magnetic Hall sensors can operate in the *voltage-mode* or the *current-mode*. In voltage-mode, the sensor is biased by a voltage ( $V_{bias}$ ) or a current ( $I_{bias}$ ) and measures the magnetic field by developing a voltage ( $V_{Hall}$ ) across the outputs. The Hall voltage with a current biasing the sensor is given by

$$V_{Hall} = G_H \frac{r_H}{n \cdot e \cdot t} B I_{bias} \quad (1)$$

where  $n$  is the doping,  $r_H$  the Hall factor,  $G_H$  the geometrical correction factor of Hall voltage,  $e$  the elementary carrier charge,  $B$  external magnetic field and  $t$  is thickness of the sensor plate. If the bias quantity is voltage, since the sensor across the two main terminal is equivalent to a resistor, we can use the law  $V = R_{in} I$ , the Hall voltage results [1]:

$$V_{Hall} \approx \mu_H G_H \frac{w}{l} B V_{bias} \quad (2)$$

here,  $\mu_H$  and  $\frac{w}{l}$  denote the Hall mobility of majority carriers and width-to-length ratio of the sensor plate, supposed to have a rectangular shape.

For the current-mode operation a current biases the sensor and the sensing terminals are kept at the same voltage. When

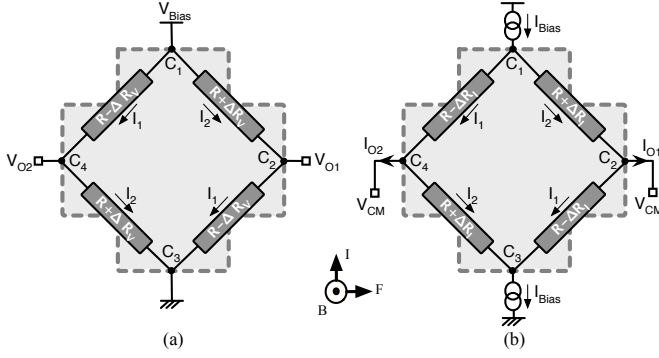


Fig. 2. Wheatstone bridge model of cross-shape Hall sensor in (a) voltage-mode and (b) current-mode configuration.

exposed to an external magnetic field, a differential current appears on the sensing terminals. This current is called the *Hall current*, given by

$$I_{Hall} = \mu_H \frac{w}{l} B I_{bias} \quad (3)$$

The above equation can be used to define the sensor sensitivity: it is the capability of the sensor to translate the magnetic field into an Hall voltage or an Hall current. We have two sensitivities  $S_{A_V}$  and  $S_{A_I}$  depending whether the output quantity is voltage or current

$$S_{A_V} = \frac{V_{Hall}}{B} \quad [V/T] \quad (4)$$

$$S_{A_I} = \frac{I_{Hall}}{B} \quad [A/T] \quad (5)$$

We can also define the relative sensitivity by dividing the above expressions by the bias quantity. For the voltage-mode we can divide equation (4) by the bias current or the applied voltage. Table I shows the sensitivity equations for the two types of Hall sensors. The dimensions are  $VA^{-1}T^{-1}$  (volt/(ampere Tesla)) and  $(V/(VT)) = T^{-1}$ .

The voltage-mode sensitivities are linked by

TABLE I  
SENSITIVITY OF HORIZONTAL HALL MAGNETIC SENSORS IN VOLTAGE-MODE AND CURRENT-MODE CONDITIONS

| Mode                        | Sensitivity                                                   | Unit              |
|-----------------------------|---------------------------------------------------------------|-------------------|
| Current-Biased Voltage-Mode | $S_{V_I} = \left  \frac{V_{Hall}}{I_{bias} \times B} \right $ | $[V/AT]$          |
| Voltage-Biased Voltage-Mode | $S_{V_V} = \left  \frac{V_{Hall}}{V_{bias} \times B} \right $ | $[V/VT] = T^{-1}$ |
| Current-Mode                | $S_I = \left  \frac{I_{Hall}}{I_{bias} \times B} \right $     | $[A/AT] = T^{-1}$ |

$$S_{V_V} = \frac{S_{V_I}}{R_{in}} \quad (6)$$

### III. PERFORMANCE COMPARISON: FIRST APPROXIMATION ANALYSIS

The applied magnetic field alters the current flow in the body sensor and this gives rise to a variation of the sheet to body resistance in the device. At the macro level we can model the sensor as a bridge of resistances as shown in Fig. 2. Four resistors model the DC electrical relationship between the input and output nodes. The applied magnetic field changes the value of resistors by  $\Delta R_V$ , as shown in the figure.

Since the voltage-mode and the current-mode of operation define different boundary conditions, the magnetic field would differently affect the current flow in the device and, therefore, the bridge resistors variation. The study requires 2D simulations. Here, in order to obtain first approximation results we suppose the same resistor change in both modes of operation.

For the voltage-mode case (Fig. 2-a) it results

$$V_{O1} = \frac{V_{bias}}{2R} (R - \Delta R_V) \quad (7)$$

$$V_{O2} = \frac{V_{bias}}{2R} (R + \Delta R_V) \quad (8)$$

The differential Hall voltage is given by

$$V_{Hall} = V_{bias} \frac{\Delta R_V}{R} \quad (9)$$

For the current-mode Hall sensors (Fig. 2-b) a bias current is injected into a terminal ( $C_1$ ) of the Hall plate, and the same bias current is draw from opposite terminal ( $C_3$ ). The output terminals ( $C_2$  and  $C_4$ ) are at the same ground voltage. The differential output currents are given by

$$I_1 = \frac{V_{C1}}{(R - \Delta R_I)} \quad (10)$$

$$I_2 = \frac{V_{C1}}{(R + \Delta R_I)} \quad (11)$$

that yield the differential Hall current as

$$I_{Hall} = I_{bias} \cdot 2 \cdot \frac{\Delta R_I}{R} \quad (12)$$

Therefore, the sensitivity in the current-mode with equal change of the resistors in the model is twice larger than the voltage-mode. There is another side benefit in having current as output variable. Its integration over a capacitance for a defined period of time determine an output voltage with a gain factor proportional to the integration time. This is a simple way to amplify the signal by using a single fully differential operational amplifier.

A possible mismatch of the voltages of the output nodes causes an offset in the output signal. This limit can be compensated for by a suitable control in the electronic interface.

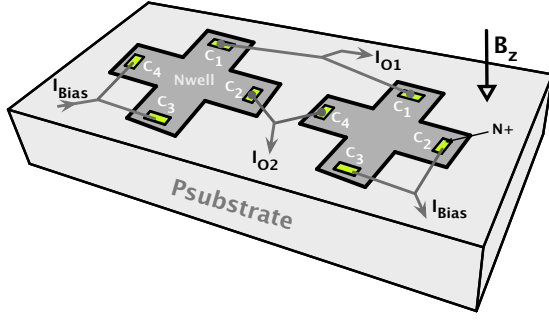


Fig. 3. Twin horizontal current-mode Hall device in CMOS technology.

#### IV. HORIZONTAL MAGNETIC HALL SENSORS

The benefit estimated in the previous section can be augmented or reduced depending on the real variation of  $\Delta R_V$  in the two modes of operation. Moreover, suitably changing the boundary conditions can further augment the sensitivity. In order to obtain an accurate comparison several current-mode horizontal and vertical Hall devices have been modelled and compared using the 2D simulator COMSOL.

The horizontal sensor of Fig. 1(b) operated in the current-mode has been studied in [2], [3] and [4]. The obtained sensitivity for doping and carrier mobility fitting a conventional 180 nm CMOS technology is  $S_I = 3.5 \%T^{-1}$ , 24% larger than the voltage-mode counterpart. The improvement depends on the different boundary conditions established by the output terminal voltage. Other physical configurations and mode of operation can further increase the sensitivity. How to find more effective solutions is difficult to understand. Because of this a systematic study of possible configurations has been performed. The solution shown in Fig. 3 depicts a twin cross-shaped sensor: the current enters into two shorted terminal of a cross shaped sensor and exits from two shorted terminals of the second sensor. The solution is suitable for the current spinning technique, normally used for compensating for the static mismatched of the device.

The COMSOL simulation with same doping and carrier mobility of the previous study give rise to the result of Fig. 4.

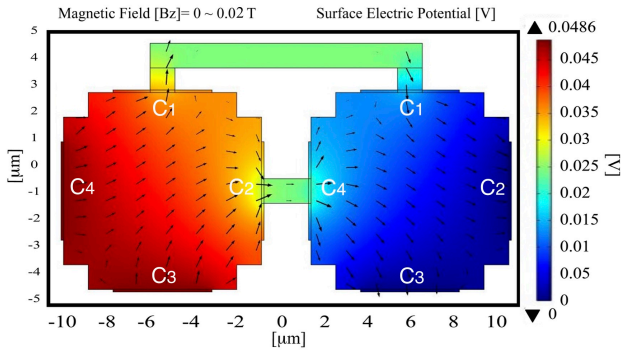


Fig. 4. COMSOL simulation of twin coupled cross-shape Horizontal Hall effect device.

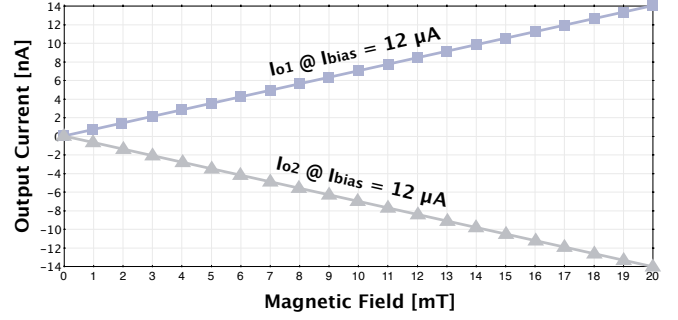


Fig. 5. Simulated output currents ( $I_{O1}$  and  $I_{O2}$ ) of a cross-shaped current-mode horizontal Hall sensor as a function of the magnetic field at 12  $\mu A$  bias current.

TABLE II  
SENSITIVITY COMPARISON OF HORIZONTAL HALL MAGNETIC SENSORS

| Reference        | Mode         | Sensor Plate    | Sensitivity [ $\%T^{-1}$ ] |
|------------------|--------------|-----------------|----------------------------|
| [5]              | Voltage-Mode | Five Strips     | 4.6                        |
| [6]              | Voltage-Mode | N-well and MOS  | 3                          |
| [7]              | Voltage-Mode | N-well Resistor | 2.5                        |
| [8]              | Voltage-Mode | N-Well          | 2.9                        |
| [9]              | Voltage-Mode | Octagonal       | 4.5                        |
| <b>This Work</b> | Current-Mode | Cross-Shape     | 11                         |

The figure shows the simulated surface electrical distribution and the current distribution when a perpendicular magnetic field of 20 mT is applied. The plot is for a bias current of 12  $\mu A$  injected in terminals  $C_3$  and  $C_4$  of the first plate and drained from terminals  $C_2$  and  $C_3$  of second plate.

Fig. 5 gives the output currents,  $I_{O1}$  and  $I_{O2}$ , as a function of magnetic field at 12  $\mu A$ . According to  $I_{Hall} = I_{O1} - I_{O2}$  and Table (I) the sensitivity can be calculated as 11  $\%T^{-1}$ . The result is 3.1 times higher than the single cross-shaped sensor operating in the current-mode.

The obtained result is more than twice published results. Even if the sensitivity depends on technology and layout, Table II compares the obtained sensitivities of various sensors operated in the voltage-mode. The twin cross-shaped coupled Horizontal Hall device in current-mode is better than other, [5], [6], [7], [8] and [9], by a factor ranging from 2.5 to 4.5.

#### V. VERTICAL MAGNETIC HALL SENSORS

Vertical magnetic sensors with structures similar to the cross section of Fig. 6 have been proposed in the literature [10], [11], [12], [13] and [14]. They operate in the voltage-mode and have sensitivity ranging from 1.5 to 4  $\%T^{-1}$ . The two-dimensional COMSOL simulation show that even for vertical sensors the sensitivity improves. We have examined the four-folded 3-contact vertical Hall device, shown in Fig. 6, and performed simulations in the voltage-mode and current-mode. Physical parameters fit a conventional 180 nm CMOS technology.

Fig. 7 shows the model geometry and the surface electrical distribution of the vertical Hall sensor when a large magnetic

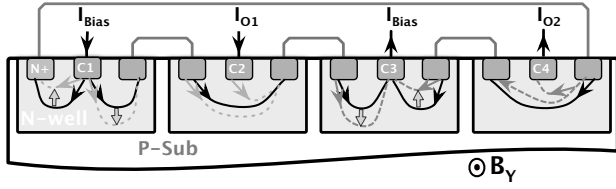


Fig. 6. Vertical 3-contact four-folded current-mode Hall device cross-section in CMOS technology.

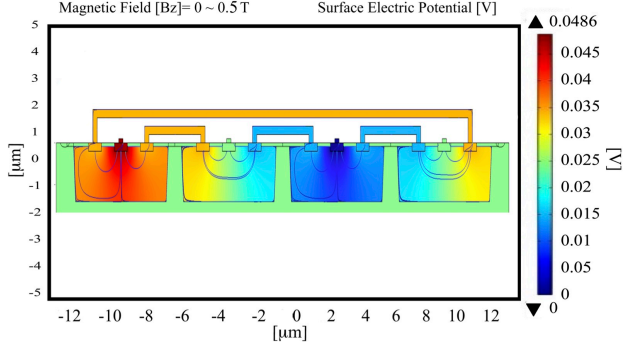


Fig. 7. COMSOL simulation of vertical 3-contact four-folded current-mode Hall device.

field (0.5 T) and a nominal bias current of 100  $\mu\text{A}$  are applied. The differential output current (Hall current) shows a good linearity at different bias currents (better than 0.2%). The output is almost 8  $\mu\text{A}$  for a magnetic field equal to 0.5 T, as shown in Fig 8. The sensitivity is as large as 16  $\%T^{-1}$ .

Table III compares published sensitivities of vertical Hall devices operated in the voltage-mode with this four-folded 3-contact Hall device. The current-mode device achieves 4 times more sensitivity than other [11], [12], [13] and [14].

## VI. CONCLUSION

This paper compares the performance of horizontal and vertical Hall sensors in voltage-mode and current-mode conditions. The simulation results focus on sensitivity. The results identify the proposed horizontal and vertical Hall sensors as

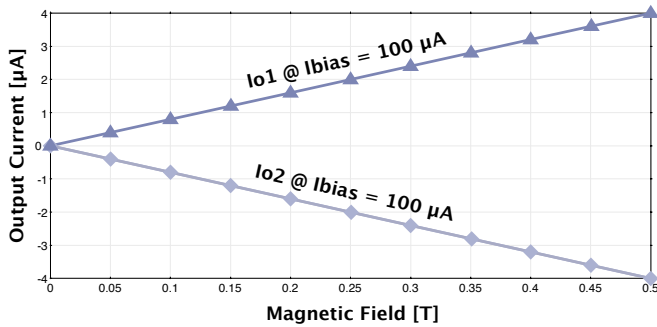


Fig. 8. Simulated output currents ( $I_{O1}$  and  $I_{O2}$ ) of vertical 3-contact four-folded current-mode Hall device as a function of the magnetic field at 100  $\mu\text{A}$  bias current.

TABLE III  
SENSITIVITY COMPARISON OF VERTICAL HALL MAGNETIC SENSORS

| Reference        | Mode         | Sensor Plate   | Sensitivity [ $\%T^{-1}$ ] |
|------------------|--------------|----------------|----------------------------|
| [11]             | Voltage-Mode | 5-Contact      | 1.52                       |
| [12]             | Voltage-Mode | 3C Four-Folded | 1.72                       |
| [13]             | Voltage-Mode | 2D 5-Contact   | 4.3                        |
| [14]             | Voltage-Mode | 5-Contact      | 4                          |
| <b>This Work</b> | Current-Mode | 3C Four-Folded | 16                         |

the optimum solution. 2D simulations using FEM COMSOL with physical parameters taken from a commercial 180nm CMOS technology permitted a comparison of various solutions. This helped us to optimize geometries and sizes. The optimum current-mode sensitivity of 11  $\%T^{-1}$  and 16  $\%T^{-1}$  have been obtained for a horizontal and vertical Hall sensor, respectively.

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